

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/341752233>

Association of Beef And Beef-Related Nutrients to Visual Cognitive Performance Training In Young Adult Females

Article in *Current Developments in Nutrition* · June 2020

DOI: 10.1093/cdn/nzaa057_049

CITATIONS

0

READS

88

6 authors, including:



Steven Riechman

Texas A&M University

152 PUBLICATIONS 2,171 CITATIONS

[SEE PROFILE](#)



Nicos Georgiades

Texas A&M University

10 PUBLICATIONS 1 CITATION

[SEE PROFILE](#)



Susanne U Mertens-Talcott

Texas A&M University

176 PUBLICATIONS 6,465 CITATIONS

[SEE PROFILE](#)



Stephen B Smith

Texas A&M University

320 PUBLICATIONS 12,243 CITATIONS

[SEE PROFILE](#)

Association of Beef And Beef-Related Nutrients to Visual Cognitive Performance Training In Young Adult Females

Steve Riechman,¹ Nicos Georgiades,¹ Caroline Sullivan,¹ Susanne Talcott,² Stephen Smith,¹ and Karen Beathard¹

¹Texas A&M University; and ²Department of Food Science at Texas A&M University

Objectives: Visual cognitive performance (VCP) is an important factor in high-speed interactions with the environment such as sports or driving, however, the testing, training and determination of factors that influence this ability has been elusive. The overall purpose of the Nutrition, Vision and Cognition in Sport Studies (IONSport) is to determine the nutritional factors that explain large variance in VCP and training responses. The objective of these studies is to determine if beef associated nutrients affect VCP.

Methods: In the preliminary study, 52 college age women performed 15 visual cognitive training sessions (Neurotracker) in 10 visits to the

laboratory (all 10 visits within 15 days). On the days of training, all food intake was documented (no intervention, no change in diet). Mean nutritional intakes were calculated for the 10 days of food records. For the RCT intervention, 80 women were randomized to either a daily beef or veggie patty and consumed 1 patty/day for 30 days.

Results: In the preliminary study, iron intake was significantly associated to VCP ($P < 0.001$) where groups of low (<10 mg/day), medium (10–20 mg/day) and high (>20 mg/day) were all significantly different from each other ($P < 0.05$). Cholesterol, choline, arginine and B vitamins intake were also significantly associated to VCP ($P < 0.05$). Preliminary analyses of the RCT to enrich these nutrients with beef indicate higher mean VCP for the beef group with significant differences in mean daily intakes of beef associated nutrients.

Conclusions: These results suggest that high intake of beef associated nutrients may improve VCP.

Funding Sources: RCT funded by National Cattleman's Beef Association.